留数法:

$$X(z)=\frac{2z^2-3z+1}{z^2-4z-5}=\frac{(2z-1)(z-1)}{(z-5)(z+1)},\quad |z|>5$$

分类讨论

※ $n=0$, $X(z)\cdot z^{n-1}=X(z)\cdot z^{-1}=\frac{(2z-1)(z-1)}{(z-5)(z+1)z}$

3个极点: $5, -1, 0$

$$\mathrm{Res}[X(z)z^{-1}]_{z=5}=\frac{(2z-1)(z-1)}{(z+1)\cdot z}=\frac{3\cancel{9}\times\cancel{4}2}{\cancel{3}\cancel{6}\times 5}=\frac{6}{5}$$

$$\mathrm{Res}[X(z)\cdot z^{-1}]_{z=-1}=\frac{(2z-1)(z-1)}{(z-5)z}=\frac{-3\times-2}{-6\times-1}=1$$

$$\mathrm{Res}[X(z)\cdot z^{-1}]_{z=0}=\frac{-1\times-1}{-5\times1}=-\frac{1}{5}$$

then, $n=0$, $x_1(n)=(\frac{6}{5}+1-\frac{1}{5})\delta(n)=2\delta(n)$

※ $n\geq1$, $\cdots$ $x_2(n)=[\frac{6}{5}(5)^n+(-1)^n]\cdot u(n-1)$

$\therefore$ $x(n)=x_1(n)+x_2(n)$

$$=2\delta(n)+[\frac{6}{5}(5)^n+(-1)^n]\cdot u(n-1)$$

$$=2\delta(n)+[\frac{6}{5}(5^n)+(-1)^n]\cdot u(n)-[\frac{6}{5}+1]\delta(n)$$

$$=-\frac{1}{5}\delta(n)+[\frac{6}{5}(5^n)+(-1)^n]\cdot u(n).$$